

Package ‘dilutionrisk’

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Type Package

Title Risk Assessment and Probability Estimation for Dilution Testing in Microbiology

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Description Comprehensive risk assessment toolkit for microbiological dilution testing in food safety. Implements probability estimation and visualization methods using bounded distributions (truncated Poisson and truncated Poisson-lognormal) for both homogeneous and heterogeneous contamination scenarios. Core functionality includes:

```
\describe{
  \item{Probability of Detection (PD)}{Estimate the likelihood
    of detecting microbial contamination exceeding specification limits
    under different dilution schemes}
  \item{Probability of Acceptance (PA)}{Evaluate sampling plan
    effectiveness using binomial acceptance probabilities}
  \item{Operating Characteristic (OC) Curves}{Visually compare
    dilution scheme performance across contamination levels}
  \item{Coefficient of Variation (CV)}{Quantify variability in
    microbial concentration estimates}
  \item{True Concentration Estimation}{Back-calculate true
    microbial concentration from observed plate counts}
  \item{Validation Framework}{Compare theoretical and
    simulation-based results to validate risk assessments}
}
```

Designed for food safety professionals, microbiologists, and researchers involved in designing and implementing microbiological monitoring programs and sampling plans compliant with food safety standards.

License GPL (>= 2)

URL <https://github.com/Mayooran1987/dilutionrisk>

BugReports <https://github.com/Mayooran1987/dilutionrisk/issues>

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 compare_cv_heterogeneous

Helper function to understand CV behavior

Description

Compares CV for different dilution schemes and parameters. Useful for sensitivity analysis.

Usage

```
compare_cv_heterogeneous(
  mu_seq,
  sd_values,
  a,
  b,
  f,
  u,
  USL,
  n_sim = 10000,
  ...
)
```

Arguments

mu_seq	Vector of mu values to evaluate.
sd_values	Vector of sd values to evaluate.
a	Lower bound.
b	Upper bound.
f	Dilution factor.
u	Plate amount.
USL	Upper specification limit.
n_sim	Number of simulations.
...	Additional arguments passed to cv_heterogeneous.

Value

A data frame with CV values.

Examples

```
## Not run:
# Compare CV across different mu and sd values
cv_sensitivity <- compare_cv_heterogeneous(
  mu_seq = seq(0, 5, by = 0.5),
  sd_values = c(0.1, 0.3, 0.5),
  a = 0, b = 300,
  f = 0.01, u = 0.1,
  USL = 1000, n_sim = 10000
)
```

```
print(cv_sensitivity)

## End(Not run)
```

cv_heterogeneous

Coefficient of Variation for Heterogeneous Batches

Description

Estimates the coefficient of variation (CV) in the original sample when diluted samples are collected from a heterogeneous (non-homogeneous) batch. The CV measures the relative variability of the estimated microbial concentration.

Usage

```
cv_heterogeneous(mu, sd, a, b, f, u, USL, n_sim)

cv_heterogeneous_multiple(mu, sd, a, b, f, u, USL, n_sim)

cv_curves_heterogeneous(mu_low, mu_high, sd, a, b, f, u, USL, n_sim)
```

Arguments

mu	Mean microbial count on the log scale. Must satisfy $\log(a) - \log(f \cdot u) \leq \mu \leq \log(b) - \log(f \cdot u)$.
sd	Standard deviation of the normal distribution on the log scale. Must be positive.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Final dilution factor. Must be between 0 and 1.
u	Amount placed on the plate. Must be positive.
USL	Upper specification limit for microbial count.
n_sim	Number of simulations. Larger values provide more precise estimates. Recommended minimum: 10,000.

Details

The coefficient of variation is defined as:

$$CV = \frac{\sigma}{\mu}$$

where σ is the standard deviation and μ is the mean of the estimated microbial concentration.

For heterogeneous batches, the microbial count follows a truncated Poisson-lognormal distribution, accounting for variability in contamination levels across the batch.

Value

Numeric value representing the coefficient of variation for the heterogeneous batch.

References

- Gonzales-Barron, U.A., Pilão Cadavez, V.A., Butler, F., 2013. Statistical approaches for the design of sampling plans for microbiological monitoring of foods. In: Mathematical and Statistical Methods in Food Science and Technology. Wiley, pp. 363–384.
- Schothorst, M. van, Zwietering, M.H., Ross, T., Buchanan, R.L., Cole, M.B., 2009. Relating microbiological criteria to food safety objectives and performance objectives. Food Control 20, 967–979.

Examples

```
# Basic usage - estimate CV for a single dilution scheme
cv_heterogeneous(mu = 2, sd = 0.2, a = 0, b = 300,
                  f = 0.01, u = 0.1, USL = 1000, n_sim = 1000)

# Compare multiple dilution schemes
cv_heterogeneous_multiple(mu = 2, sd = 0.2, a = 0, b = 300,
                           f = c(0.01, 0.1), u = c(0.1, 0.1),
                           USL = 1000, n_sim = 1000)

# Generate CV curves across a range of mean microbial counts
cv_curves_heterogeneous(mu_low = -5, mu_high = 10, sd = 0.2,
                         a = 0, b = 300, f = c(0.01, 0.1),
                         u = c(0.1, 0.1), USL = 1000, n_sim = 1000)

# Quick example with small simulation size for testing
cv_heterogeneous(mu = 2, sd = 0.2, a = 0, b = 300,
                  f = 0.01, u = 0.1, USL = 1000, n_sim = 500)
```

cv_homogeneous

Coefficient of Variation for Homogeneous Batches

Description

Estimates the coefficient of variation (CV) in the original sample when diluted samples are collected from a homogeneous batch.

Usage

```
cv_homogeneous(lambda, a, b, f, u, USL, n_sim)

cv_homogeneous_multiple(lambda, a, b, f, u, USL, n_sim)

cv_curves_homogeneous(lambda_low, lambda_high, a, b, f, u, USL, n_sim)
```

Arguments

lambda	Expected microbial count (λ). Must be non-negative.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.

f	Final dilution factor. Must be between 0 and 1.
u	Amount placed on the plate. Must be positive.
USL	Upper specification limit for microbial count.
n_sim	Number of simulations. Larger values provide more precise estimates. Recommended minimum: 10,000.
lambda_low	Lower bound of expected microbial count for x-axis in graphical displays.
lambda_high	Upper bound of expected microbial count for x-axis in graphical displays.

Value

Numeric value representing the coefficient of variation for the homogeneous batch.

Examples

```
# Basic usage
cv_homogeneous(lambda = 2000, a = 0, b = 300,
                f = 0.01, u = 0.1, USL = 1000, n_sim = 5000)
```

OC_curves_heterogeneous

Operating Characteristic (OC) Curves for Heterogeneous Batches

Description

Generates operating characteristic curves comparing different dilution schemes when samples are collected from a heterogeneous (non-homogeneous) batch.

Usage

```
OC_curves_heterogeneous(
  c,
  mu_low,
  mu_high,
  sd,
  a,
  b,
  f,
  u,
  USL,
  n,
  type = "theory",
  n_sim = NA
)
```

Arguments

c	Acceptance number (maximum number of defective units allowed).
mu_low	Lower bound of mean microbial count for x-axis (log scale).
mu_high	Upper bound of mean microbial count for x-axis (log scale).

sd	Standard deviation of the normal distribution on the log scale.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Vector of final dilution factors.
u	Vector of amounts placed on the plate.
USL	Upper specification limit for microbial count.
n	Number of samples inspected. Must be positive integer.
type	Type of calculation: "theory" (default) or "simulation".
n_sim	Number of simulations. Required when type = "simulation".

Value

A ggplot object showing OC curves for different dilution schemes.

Examples

```
OC_curves_heterogeneous(c = 2, mu_low = 4, mu_high = 9, sd = 0.2,
  a = 0, b = 300, f = c(0.01, 0.1), u = c(0.1, 0.1),
  USL = 1000, n = 5)
```

OC_curves_homogeneous *Operating Characteristic (OC) Curves for Homogeneous Batches*

Description

Generates operating characteristic curves comparing different dilution schemes when samples are collected from a homogeneous batch.

Usage

```
OC_curves_homogeneous(
  c,
  lambda_low,
  lambda_high,
  a,
  b,
  f,
  u,
  USL,
  n,
  type = "theory",
  n_sim = NA
)
```

Arguments

c	Acceptance number (maximum number of defective units allowed).
lambda_low	Lower bound of expected microbial count for x-axis.
lambda_high	Upper bound of expected microbial count for x-axis.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Vector of final dilution factors.
u	Vector of amounts placed on the plate.
USL	Upper specification limit for microbial count.
n	Number of samples inspected. Must be positive integer.
type	Type of calculation: "theory" (default) or "simulation".
n_sim	Number of simulations. Required when type = "simulation".

Details

Operating characteristic curves show the probability of acceptance as a function of the true microbial concentration. They are useful for comparing the performance of different dilution schemes and sampling plans.

Value

A ggplot object showing OC curves for different dilution schemes.

Examples

```
OC_curves_homogeneous(c = 2, lambda_low = 0, lambda_high = 5000,
  a = 0, b = 300, f = c(0.01, 0.1), u = c(0.1, 0.1),
  USL = 1000, n = 5)
```

pd_curves_heterogeneous

Probability of Detection Curves for Heterogeneous Batches

Description

Generates probability of detection curves comparing different dilution schemes when samples are collected from a heterogeneous (non-homogeneous) batch.

Usage

```
pd_curves_heterogeneous(
  mu_low,
  mu_high,
  sd,
  a,
  b,
```



```

    f,
    u,
    USL,
    type = "theory",
    n_sim = NA
  )

```

Arguments

mu_low	Lower bound of mean microbial count for x-axis (log scale).
mu_high	Upper bound of mean microbial count for x-axis (log scale).
sd	Standard deviation of the normal distribution on the log scale.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Vector of final dilution factors.
u	Vector of amounts placed on the plate.
USL	Upper specification limit for microbial count.
type	Type of calculation: "theory" (default) or "simulation".
n_sim	Number of simulations. Required when type = "simulation".

Value

A ggplot object showing PD curves for different dilution schemes.

Examples

```

pd_curves_heterogeneous(mu_low = 0, mu_high = 10, sd = 0.2,
                        a = 0, b = 300, f = c(0.01, 0.1), u = c(0.1, 0.1),
                        USL = 1000)

```

pd_curves_homogeneous *Probability of Detection Curves for Homogeneous Batches*

Description

Generates probability of detection curves comparing different dilution schemes when samples are collected from a homogeneous batch.

Usage

```

pd_curves_homogeneous(
  lambda_low,
  lambda_high,
  a,
  b,
  f,
  u,
  USL,
  type = "theory",
  n_sim = NA
)

```

Arguments

lambda_low	Lower bound of expected microbial count for x-axis.
lambda_high	Upper bound of expected microbial count for x-axis.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Vector of final dilution factors.
u	Vector of amounts placed on the plate.
USL	Upper specification limit for microbial count.
type	Type of calculation: "theory" (default) or "simulation".
n_sim	Number of simulations. Required when type = "simulation".

Value

A ggplot object showing PD curves for different dilution schemes.

Examples

```
pd_curves_homogeneous(lambda_low = 0, lambda_high = 5000,
                        a = 0, b = 300, f = c(0.01, 0.1), u = c(0.1, 0.1),
                        USL = 1000)
```

pd_validation_heterogeneous

Comparison based on probability of detection curves for different dilution schemes when diluted samples collected from a heterogeneous batch.

Description

`pd_validation_heterogeneous` provides the probability of detection curves for validate the results when samples collected from a heterogeneous batch.

Usage

```
pd_validation_heterogeneous(mu_low, mu_high, sd, a, b, f, u, USL, n_sim)
```

Arguments

mu_low	the lower value of the mean microbial count (μ) for use in the graphical display's x-axis (on the log scale).
mu_high	the upper value of the mean microbial count (μ) for use in the graphical display's x-axis (on the log scale).
sd	the standard deviation of the normal distribution (on the log scale).
a	lower domain of the number of cell counts.
b	upper domain of the number of cell counts.
f	final dilution factor.
u	amount put on the plate.
USL	upper specification limit.
n_sim	number of simulations (large simulations provide more precise estimations).

Details

[pd_curves_heterogeneous](#) provides probability of detection curves for different dilution schemes when samples collected from a heterogeneous batch (this section will be updated later on).

Value

Probability of detection curves when diluted samples collected from a heterogeneous batch.

Examples

```
mu_low <- 1
mu_high <- 12
sd <- 0.2
a <- 0
b <- 300
f <- 0.01
u <- 0.1
USL <- 1000
n_sim <- 5000
pd_validation_heterogeneous(mu_low, mu_high, sd, a, b, f, u, USL, n_sim)
```

pd_validation_homogeneous

Comparison based on probability of detection curves for different dilution schemes when diluted samples collected from a homogeneous batch.

Description

[pd_validation_homogeneous](#) provides the probability of detection curves for validate the results when samples collected from a homogeneous batch.

Usage

```
pd_validation_homogeneous(lambda_low, lambda_high, a, b, f, u, USL, n_sim)
```

Arguments

lambda_low	the lower value of the expected microbial count (λ) for use in the graphical display's x-axis.
lambda_high	the upper value of the expected microbial count (λ) for use in the graphical display's x-axis.
a	lower domain of the number of microbial count.
b	upper domain of the number of microbial count.
f	final dilution factor.
u	amount put on the plate.
USL	upper specification limit.
n_sim	number of simulations (large simulations provide more precise estimations).

Details

[pd_curves_homogeneous](#) provides probability of detection curves for different dilution schemes when samples collected from a homogeneous batch (this section will be updated later on).

Value

Probability of detection curves when diluted samples collected from a homogeneous batch.

Examples

```
lambda_low <- 0
lambda_high <- 5000
a <- 0
b <- 300
f <- 0.01
u <- 0.1
USL <- 1000
n_sim <- 5000
pd_validation_homogeneous(lambda_low, lambda_high, a, b, f, u, USL, n_sim = 500)
pd_validation_homogeneous(lambda_low, lambda_high, a, b, f, u, USL, n_sim = 5000)
```

plot_cv_comparison	<i>Visualize CV Comparison</i>
--------------------	--------------------------------

Description

Creates a heatmap or contour plot of CV values across mu and sd parameters.

Usage

```
plot_cv_comparison(cv_data, type = "heatmap")
```

Arguments

cv_data	Data frame from <code>compare_cv_heterogeneous</code> .
type	Type of plot: "heatmap" or "contour".

Value

A ggplot object.

Examples

```
cv_data <- compare_cv_heterogeneous(
  mu_seq = seq(0, 5, by = 0.25),
  sd_values = seq(0.1, 0.8, by = 0.05),
  a = 0, b = 300,
  f = 0.01, u = 0.1,
  USL = 1000, n_sim = 5000
)
plot_cv_comparison(cv_data, type = "heatmap")
```

prob_acceptance_heterogeneous

Probability of Acceptance for Heterogeneous Batches

Description

Calculates the probability of acceptance (PA) in the original sample when diluted samples are collected from a heterogeneous (non-homogeneous) batch.

Usage

```
prob_acceptance_heterogeneous(  
  c,  
  mu,  
  sd,  
  a,  
  b,  
  f,  
  u,  
  USL,  
  n,  
  type = "theory",  
  n_sim = NA  
)  
  
prob_acceptance_heterogeneous_multiple(  
  c,  
  mu,  
  sd,  
  a,  
  b,  
  f,  
  u,  
  USL,  
  n,  
  type = "theory",  
  n_sim = NA  
)
```

Arguments

c	Acceptance number (maximum number of defective units allowed).
mu	Mean microbial count on the log scale. Must satisfy $\log(a) - \log(f \cdot u) \leq \mu \leq \log(b) - \log(f \cdot u)$.
sd	Standard deviation of the normal distribution on the log scale. Must be positive.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Final dilution factor. Must be between 0 and 1.
u	Amount placed on the plate. Must be positive.

USL	Upper specification limit for microbial count.
n	Number of samples inspected. Must be positive integer.
type	Type of calculation: "theory" (default) or "simulation".
n_sim	Number of simulations. Required when type = "simulation".

Details

The probability of acceptance is calculated using the binomial distribution:

$$P_a = P(X \leq c) = \sum_{k=0}^c \binom{n}{k} p_d^k (1 - p_d)^{n-k}$$

where p_d is the probability of detection and n is the number of samples inspected.

Value

Numeric value representing the probability of acceptance.

Examples

```
# Basic usage
prob_acceptance_heterogeneous(c = 2, mu = 7, sd = 0.2, a = 0, b = 300,
                              f = 0.01, u = 0.1, USL = 1000, n = 5)

# Multiple dilution schemes
prob_acceptance_heterogeneous_multiple(c = 2, mu = 7, sd = 0.2,
                                       a = 0, b = 300,
                                       f = c(0.01, 0.1), u = c(0.1, 0.1),
                                       USL = 1000, n = 5)
```

prob_acceptance_heterogeneous_multiple

Probability of acceptance estimation when diluted samples are collected from a heterogeneous batch.

Description

[prob_acceptance_heterogeneous_multiple](#) provides a probability of acceptance in the original sample when samples collected from a heterogeneous batch.

Usage

```
prob_acceptance_heterogeneous_multiple (c, mu, sd, a, b, f, u, USL, n, type, n_sim)
```

Arguments

c	acceptance number
mu	the mean microbial count (on the log scale).
sd	the standard deviation of the normal distribution (on the log scale).
a	lower domain of the number of cell counts.
b	upper domain of the number of cell counts.
f	final dilution factor.
u	amount put on the plate.
USL	upper specification limit.
n	number of samples which are used for inspection.
type	what type of the results you would like to consider such as "theory" or "simulation" (default "theory").
n_sim	number of simulations (large simulations provide more precise estimations).

Details

[prob_acceptance_heterogeneous_multiple](#) provides a probability of acceptance when diluted samples are collected from a heterogeneous batch (this section will be updated later on).

Value

Probability of acceptance when samples collected from a heterogeneous batch.

Examples

```
c <- 2
mu <- 7
sd <- 0.2
a <- 0
b <- 300
f <- c(0.01, 0.1, 1)
u <- c(0.1, 0.1, 0.1)
USL <- 1000
n <- 5
n_sim <- 50000
prob_acceptance_heterogeneous_multiple(c, mu, sd, a, b, f, u, USL, n)
```

prob_acceptance_homogeneous

Probability of Acceptance for Homogeneous Batches

Description

Calculates the probability of acceptance (PA) in the original sample when diluted samples are collected from a homogeneous batch.

Usage

```

prob_acceptance_homogeneous(
  c,
  lambda,
  a,
  b,
  f,
  u,
  USL,
  n,
  type = "theory",
  n_sim = NA
)

prob_acceptance_homogeneous_multiple(
  c,
  lambda,
  a,
  b,
  f,
  u,
  USL,
  n,
  type = "theory",
  n_sim = NA
)

```

Arguments

c	Acceptance number (maximum number of defective units allowed).
lambda	Expected microbial count (λ). Must be positive.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Final dilution factor. Must be between 0 and 1.
u	Amount placed on the plate. Must be positive.
USL	Upper specification limit for microbial count.
n	Number of samples inspected. Must be positive integer.
type	Type of calculation: "theory" (default) or "simulation".
n_sim	Number of simulations. Required when type = "simulation".

Details

The probability of acceptance is calculated using the binomial distribution:

$$P_a = P(X \leq c) = \sum_{k=0}^c \binom{n}{k} p_d^k (1 - p_d)^{n-k}$$

where p_d is the probability of detection and n is the number of samples inspected.

Value

Numeric value representing the probability of acceptance.

Examples

```
# Basic usage
prob_acceptance_homogeneous(c = 2, lambda = 2000, a = 0, b = 300,
                             f = 0.01, u = 0.1, USL = 1000, n = 5)

# Multiple dilution schemes
prob_acceptance_homogeneous_multiple(c = 2, lambda = 2000, a = 0, b = 300,
                                      f = c(0.01, 0.1), u = c(0.1, 0.1),
                                      USL = 1000, n = 5)
```

prob_acceptance_homogeneous_multiple

*Probability of acceptance estimation for multiple dilution schemes
when diluted samples are collected from a homogeneous batch.*

Description

[prob_acceptance_homogeneous_multiple](#) provides a probability of acceptance for multiple dilution schemes in the original sample when samples collected from a homogeneous batch

Usage

```
prob_acceptance_homogeneous_multiple(c, lambda, a, b, f, u, USL, n, type, n_sim)
```

Arguments

c	acceptance number
lambda	the expected microbial count (λ).
a	lower domain of the number of microbial count.
b	upper domain of the number of microbial count.
f	final dilution factor.
u	amount put on the plate.
USL	upper specification limit.
n	number of samples which are used for inspection.
type	what type of the results you would like to consider such as "theory" or "simulation" (default "theory").
n_sim	number of simulations (large simulations provide more precise estimations).

Details

[prob_detection_homogeneous_multiple](#) provides a probability of acceptance for multiple dilution schemes in the original sample when samples collected from a homogeneous batch (this section will be updated later on).

Value

Probability of acceptance when diluted samples are collected from a homogeneous batch.

Examples

```
c <- 2
lambda <- 1000
a <- 0
b <- 300
f <- c(0.01, 0.1, 1)
u <- c(0.1, 0.1, 0.1)
USL <- 1000
n <- 5
n_sim <- 50000
prob_acceptance_homogeneous_multiple(c, lambda, a, b, f, u, USL, n)
```

prob_detection_heterogeneous

Probability of Detection for Heterogeneous Batches

Description

Calculates the probability of detection (PD) in the original sample when diluted samples are collected from a heterogeneous (non-homogeneous) batch.

Usage

```
prob_detection_heterogeneous(
  mu,
  sd,
  a,
  b,
  f,
  u,
  USL,
  type = "theory",
  n_sim = NA
)

prob_detection_heterogeneous_multiple(
  mu,
  sd,
  a,
  b,
  f,
  u,
  USL,
  type = "theory",
  n_sim = NA
)
```

prob_detection_heterogeneous_multiple

Probability of detection estimation for multiple dilution schemes when diluted samples are collected from a heterogeneous batch.

Description

`prob_detection_heterogeneous_multiple` provides a probability of detection for multiple dilution schemes in the original sample when samples collected from a heterogeneous batch.

Usage

```
prob_detection_heterogeneous_multiple(mu, sd, a, b, f, u, USL, type, n_sim)
```

Arguments

<code>mu</code>	the mean microbial count (on the log scale).
<code>sd</code>	the standard deviation of the normal distribution (on the log scale).
<code>a</code>	lower domain of the number of cell counts.
<code>b</code>	upper domain of the number of cell counts.
<code>f</code>	vector of final dilution factor.
<code>u</code>	vector of amount put on the plate.
<code>USL</code>	upper specification limit.
<code>type</code>	what type of the results you would like to consider such as "theory" or "simulation" (default "theory").
<code>n_sim</code>	number of simulations (large simulations provide a more precise estimation).

Details

`prob_detection_heterogeneous_multiple` provides a probability of detection when diluted samples are collected from a heterogeneous batch. We define the random variable X_i is the number of colonies on the i^{th} plate. In practice, the acceptance for countable numbers of colonies on a plate must be between 30 and 300. Therefore, we can utilise bounded distributions to model the number of colonies on a plate. In the heterogeneous case, we employed truncated Poisson lognormal distribution to the model. (this section will be updated later on).

Value

Probability of detection when samples collected from a heterogeneous batch.

Examples

```
mu <- 7
sd <- 0.2
a <- 0
b <- 300
f <- c(0.01, 0.1, 1)
u <- c(0.1, 0.1, 0.1)
USL <- 1000
n_sim <- 50000
prob_detection_heterogeneous_multiple(mu, sd, a, b, f, u, USL, n_sim)
```

prob_detection_homogeneous

Probability of Detection for Homogeneous Batches

Description

Calculates the probability of detection (PD) in the original sample when diluted samples are collected from a homogeneous batch.

Usage

```
prob_detection_homogeneous(
  lambda,
  a,
  b,
  f,
  u,
  USL,
  type = "theory",
  n_sim = NA
)

prob_detection_homogeneous_multiple(
  lambda,
  a,
  b,
  f,
  u,
  USL,
  type = "theory",
  n_sim = NA
)
```

Arguments

lambda	Expected microbial count (λ). Must be positive.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Final dilution factor. Must be between 0 and 1.
u	Amount placed on the plate. Must be positive.
USL	Upper specification limit for microbial count.
type	Type of calculation: "theory" (default) or "simulation".
n_sim	Number of simulations. Required when type = "simulation".

Details

The probability of detection is defined as the probability that the estimated microbial count exceeds the upper specification limit (USL). For homogeneous batches, the count follows a truncated Poisson distribution.

The theoretical probability is calculated as:

$$P_d = 1 - \sum_{i=1}^{USL \times f \times u} \frac{(\lambda f u)^i}{i! (e^{\lambda f u} - 1)}$$

Value

Numeric value representing the probability of detection.

References

- Schothorst, M. van, Zwietering, M.H., Ross, T., Buchanan, R.L., Cole, M.B., 2009. Relating microbiological criteria to food safety objectives and performance objectives. Food Control 20, 967–979.

Examples

```
# Theoretical calculation
prob_detection_homogeneous(lambda = 2000, a = 0, b = 300,
                           f = 0.01, u = 0.1, USL = 1000)

# Simulation-based calculation
prob_detection_homogeneous(lambda = 2000, a = 0, b = 300,
                           f = 0.01, u = 0.1, USL = 1000,
                           type = "simulation", n_sim = 50000)
```

prob_detection_homogeneous_multiple

Probability of detection estimation for multiple dilution schemes when diluted samples are collected from a homogeneous batch.

Description

[prob_detection_homogeneous_multiple](#) provides a probability of detection for multiple dilution schemes in the original sample when samples collected from a homogeneous batch.

Usage

```
prob_detection_homogeneous_multiple(lambda, a, b, f, u, USL, type, n_sim)
```

Arguments

lambda	the expected microbial count (λ).
a	lower domain of the number of microbial count.
b	upper domain of the number of microbial count.
f	final dilution factor.
u	amount put on the plate.
USL	upper specification limit.
type	what type of the results you would like to consider such as "theory" or "simulation" (default "theory").
n_sim	number of simulations (large simulations provide more precise estimations).

Details

`prob_detection_homogeneous_multiple` provides a probability of detection when the diluted sample has homogeneous contaminants. We define the random variable X_i is the number of colonies on the i^{th} plate. In practice, the acceptance for countable numbers of colonies on a plate must be between 30 and 300. Therefore, we can utilise bounded distributions to model the number of colonies on a plate. In the homogeneous case, we employed truncated Poisson distribution to model (this section will be updated later on).

Value

Probability of detection when diluted samples are collected from a homogeneous batch.

Examples

```
lambda <- 1000
a <- 0
b <- 300
f <- c(0.01, 0.1, 1)
u <- c(0.1, 0.1, 0.1)
USL <- 1000
n_sim <- 5000
prob_detection_homogeneous_multiple(lambda, a, b, f, u, USL)
```

rt pois

Truncated Poisson Random Number Generation

Description

Generates random deviates from a truncated Poisson distribution. The function uses rejection sampling to generate observations from a Poisson distribution conditioned to lie within the interval [a](#), [b](#).

Usage

```
rt pois(n, lambda, a, b)
```

Arguments

n	Number of observations. Must be a positive integer.
lambda	Expected count parameter. Must be non-negative.
a	Lower truncation point. Must be non-negative.
b	Upper truncation point. Must be greater than a.

Details

The truncated Poisson distribution has probability mass function:

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x! \sum_{k=a}^b \frac{\lambda^k e^{-\lambda}}{k!}}$$

for $a \leq x \leq b$.

The function uses rejection sampling with a maximum of 1,000,000 iterations to ensure convergence. For $\lambda = 0$, all observations are 0 if within bounds.

Value

A vector of random numbers from the truncated Poisson distribution. Returns NA if rejection sampling fails to converge.

References

- Johnson, N. L., Kemp, A. W., & Kotz, S. (2005). *Univariate Discrete Distributions* (3rd ed.). Wiley.

Examples

```
# Generate 10 observations from truncated Poisson with lambda = 5
# bounded between 0 and 300
set.seed(123)
rtpois(n = 10, lambda = 5, a = 0, b = 300)

# Generate 5 observations with lower bound 10 and upper bound 50
rtpois(n = 5, lambda = 20, a = 10, b = 50)

# Lambda = 0 case (all zeros)
rtpois(n = 5, lambda = 0, a = 0, b = 300)

# Generate for different lambda values
lambdas <- c(1, 5, 10, 20)
set.seed(456)
for (lam in lambdas) {
  cat("Lambda =", lam, ":",
      rtpois(n = 3, lambda = lam, a = 0, b = 100), "\n")
}

# Compare with regular Poisson (when bounds are wide)
set.seed(789)
trunc_sample <- rtpois(n = 1000, lambda = 10, a = 0, b = 300)
regular_sample <- rpois(n = 1000, lambda = 10)

# Both should be similar when bounds are not restrictive
cat("Truncated mean:", mean(trunc_sample), "\n")
cat("Regular mean:", mean(regular_sample), "\n")

# When bounds are restrictive (0-10 only)
set.seed(321)
trunc_restricted <- rtpois(n = 1000, lambda = 20, a = 0, b = 10)
cat("Restricted truncated mean:", mean(trunc_restricted), "\n")

# Note: Rejection sampling may be slow for very restrictive bounds
# or very large lambda values
```


Description

Generates random deviates from a truncated Poisson-lognormal distribution. This is a compound distribution where the Poisson mean parameter follows a lognormal distribution, conditioned on the Poisson outcome being within the interval [a, b](#).

Generates random deviates from a truncated Poisson-lognormal distribution.

Usage

```
rtrunpoilog(n, mu, sd, a, b)
```

```
rtrunpoilog(n, mu, sd, a, b)
```

Arguments

n	Number of observations. If <code>length(n) > 1</code> , the length is taken to be the number required.
mu	Mean of the lognormal distribution on the log scale.
sd	Standard deviation of the lognormal distribution on the log scale.
a	Lower truncation point (lower bound of cell count).
b	Upper truncation point (upper bound of cell count).

Details

The Poisson-lognormal distribution is a mixture distribution where:

- $\Lambda \sim \text{Lognormal}(\mu, \sigma^2)$
- $X|\Lambda = \lambda \sim \text{Poisson}(\lambda)$

The truncation limits the support to $a \leq X \leq b$.

The probability mass function is given by:

$$P(X = x) = \frac{1}{x! \sqrt{2\pi}\sigma} \int_0^\infty \lambda^{x-1} e^{-\lambda} \exp\left(-\frac{(\log(\lambda) - \mu)^2}{2\sigma^2}\right) d\lambda$$

This distribution is particularly useful for modeling microbial counts in heterogeneous (non-homogeneous) batches where the contamination level varies across the batch.

The Poisson-lognormal distribution is a mixture distribution where:

- $\Lambda \sim \text{Lognormal}(\mu, \sigma^2)$
- $X|\Lambda = \lambda \sim \text{Poisson}(\lambda)$

The truncation limits the support to $a \leq X \leq b$.

Value

A vector of random numbers from the truncated Poisson-lognormal distribution. Returns NA if rejection sampling fails to converge.

A vector of random numbers from the truncated Poisson-lognormal distribution.

References

- Gonzales-Barron, U.A., Pilão Cadavez, V.A., Butler, F., 2013. Statistical approaches for the design of sampling plans for microbiological monitoring of foods. In: Mathematical and Statistical Methods in Food Science and Technology. Wiley, pp. 363–384.
- Bulmer, M. G. (1974). On fitting the Poisson lognormal distribution to species-abundance data. *Biometrics*, 30(1), 101-110.

Examples

```
# Generate 10 observations from truncated Poisson-lognormal
# with mu = 2, sd = 0.5, bounded between 0 and 300
set.seed(123)
rtrunpoilog(n = 10, mu = 2, sd = 0.5, a = 0, b = 300)

# Generate with different parameters
set.seed(456)
rtrunpoilog(n = 5, mu = 1, sd = 0.3, a = 0, b = 100)

# Compare with regular Poisson-lognormal (when bounds are wide)
set.seed(789)
n_sim <- 1000
trunc_sample <- rtrunpoilog(n = n_sim, mu = 2, sd = 0.5, a = 0, b = 300)
regular_sample <- rpois(n = n_sim, lambda = rlnorm(n_sim, 2, 0.5))

cat("Truncated mean:", mean(trunc_sample), "\n")
cat("Regular mean:", mean(regular_sample), "\n")
cat("Truncated variance:", var(trunc_sample), "\n")
cat("Regular variance:", var(regular_sample), "\n")

# When bounds are restrictive (0-10 only)
set.seed(321)
trunc_restricted <- rtrunpoilog(n = n_sim, mu = 3, sd = 0.8, a = 0, b = 10)
cat("Restricted truncated mean:", mean(trunc_restricted), "\n")

# Generate samples with different sd values
mu_val <- 2
sd_values <- c(0.2, 0.5, 1.0)
set.seed(654)
for (sd_val in sd_values) {
  sample <- rtrunpoilog(n = 10, mu = mu_val, sd = sd_val, a = 0, b = 300)
  cat("mu =", mu_val, "sd =", sd_val, ":",
      summary(sample), "\n")
}

# Note: Rejection sampling may be slow for very restrictive bounds
# or when sd is very large

# Generate 100 random values
set.seed(123)
rtrunpoilog(n = 100, mu = 0, sd = 1, a = 0, b = 300)

# Generate a single value
rtrunpoilog(n = 1, mu = 0.5, sd = 0.8, a = 10, b = 200)
```

true_concentration_heterogeneous

True Concentration Estimation for Heterogeneous Batches

Description

Estimates the true microbial concentration in the original sample when diluted samples are collected from a heterogeneous (non-homogeneous) batch.

Usage

```
true_concentration_heterogeneous(mu, sd, a, b, f, u, USL, n_sim)
```

```
true_concentration_heterogeneous_multiple(mu, sd, a, b, f, u, USL, n_sim)
```

```
true_concentration_curves_heterogeneous(
  mu_low,
  mu_high,
  sd,
  a,
  b,
  f,
  u,
  USL,
  n_sim
)
```

Arguments

mu	Mean microbial count on the log scale. Must satisfy $\log(a) - \log(f \cdot u) \leq \mu \leq \log(b) - \log(f \cdot u)$.
sd	Standard deviation of the normal distribution on the log scale.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Final dilution factor. Must be between 0 and 1.
u	Amount placed on the plate. Must be positive.
USL	Upper specification limit for microbial count.
n_sim	Number of simulations. Larger values provide more precise estimates. Recommended minimum: 10,000.
mu_low	Lower bound of mean microbial count for x-axis (log scale).
mu_high	Upper bound of mean microbial count for x-axis (log scale).

Details

The true concentration is estimated as:

$$C = \frac{X}{f \times u}$$

where X is the count of microorganisms on a plate following a truncated Poisson-lognormal distribution, f is the final dilution factor, and u is the amount placed on the plate.

Value

Numeric value representing the estimated true concentration.

References

- Gonzales-Barron, U.A., Pilão Cadavez, V.A., Butler, F., 2013. Statistical approaches for the design of sampling plans for microbiological monitoring of foods. In: Mathematical and Statistical Methods in Food Science and Technology. Wiley, pp. 363–384.

Examples

```
# Basic usage
true_concentration_heterogeneous(mu = 2, sd = 0.2, a = 0, b = 300,
                                f = 0.01, u = 0.1, USL = 1000, n_sim = 5000)

# Multiple dilution schemes
true_concentration_heterogeneous_multiple(mu = 2, sd = 0.2, a = 0, b = 300,
                                           f = c(0.01, 0.1), u = c(0.1, 0.1),
                                           USL = 1000, n_sim = 5000)

# Plot concentration curves
true_concentration_curves_heterogeneous(mu_low = 0, mu_high = 10, sd = 0.2,
                                         a = 0, b = 300, f = c(0.01, 0.1),
                                         u = c(0.1, 0.1), USL = 1000,
                                         n_sim = 5000)
```

true_concentration_homogeneous

True Concentration Estimation for Homogeneous Batches

Description

Estimates the true microbial concentration in the original sample when diluted samples are collected from a homogeneous batch.

Usage

```
true_concentration_homogeneous(lambda, a, b, f, u, USL, n_sim)

true_concentration_homogeneous_multiple(lambda, a, b, f, u, USL, n_sim)

true_concentration_curves_homogeneous(
  lambda_low,
  lambda_high,
  a,
  b,
  f,
  u,
  USL,
  n_sim
)
```

Arguments

lambda	Expected microbial count (λ). Must be positive.
a	Lower bound of cell count domain. Must be non-negative.
b	Upper bound of cell count domain. Must be greater than a.
f	Final dilution factor. Must be between 0 and 1.
u	Amount placed on the plate. Must be positive.
USL	Upper specification limit for microbial count.
n_sim	Number of simulations. Larger values provide more precise estimates. Recommended minimum: 10,000.
lambda_low	Lower bound of expected microbial count for x-axis.
lambda_high	Upper bound of expected microbial count for x-axis.

Details

The true concentration is estimated as:

$$C = \frac{X}{f \times u}$$

where X is the count of microorganisms on a plate following a truncated Poisson distribution, f is the final dilution factor, and u is the amount placed on the plate.

Value

Numeric value representing the estimated true concentration.

Examples

```
# Basic usage
true_concentration_homogeneous(lambda = 2000, a = 0, b = 300,
                                f = 0.01, u = 0.1, USL = 1000, n_sim = 5000)

# Multiple dilution schemes
true_concentration_homogeneous_multiple(lambda = 2000, a = 0, b = 300,
                                         f = c(0.01, 0.1), u = c(0.1, 0.1),
                                         USL = 1000, n_sim = 5000)

# Plot concentration curves
true_concentration_curves_homogeneous(lambda_low = 0, lambda_high = 5000,
                                       a = 0, b = 300, f = c(0.01, 0.1),
                                       u = c(0.1, 0.1), USL = 1000,
                                       n_sim = 5000)
```

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